

Data-Efficient Linear Visual State Estimation

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Abstract—Lee *et al.* [1] demonstrate that a pixel-based Luenberger observer whose gain matrix is learned from teacher demonstrations via linear least squares can achieve reliable cartpole stabilisation using only a robot-facing camera—a method they call *pixels to torques*. This work implements and evaluates a closely related hybrid observer in simulation. The observer combines nominal linearised dynamics with a Ridge regression model that maps raw binary pixel frames to pole-angle estimates, which are blended into the model prediction. Trained on 150 teacher demonstrations and evaluated on 100 random initial conditions under true plant mismatch, the observer achieves 100% survival and 79% paper-success (cart ≤ 0.176 m, pole $\leq 2^\circ$ at $t = 2.5$ s). The paper discusses how this simplified single-state-correction variant relates to the full observer of Lee *et al.* [1] and identifies cart-position estimation as the primary remaining limitation.

I. INTRODUCTION

A robotic arm operating in an unstructured environment has no intrinsic knowledge of where objects are relative to its end-effector. A camera is therefore not optional: it is a fundamental requirement for any manipulation task. Given that the camera is already present, a natural question arises: rather than building a layered perception pipeline to extract features, estimate poses, and feed a separate controller, why not learn to map directly from pixels to motor commands?

Existing approaches to vision-based control each carry significant limitations. **Reinforcement learning (RL)** policies can handle rich dynamics but require high-fidelity simulation to generate training data. The resulting sim-to-real gap degrades real-world performance, and closing it through domain randomisation leaves policy performance on the table [3]. **End-to-end vision-based RL** removes the perception module entirely by learning a neural-network policy from pixels to actions. However, it demands orders of magnitude more data than model-based methods, and the resulting network is often too compute-intensive to run at the high control frequencies required for dynamic tasks [4].

Linear vision-based control [1] resolves both limitations. A Luenberger observer [5] is learned directly on hardware via linear regression, using demonstration data from a privileged teacher controller with access to ground-truth sensor readings. The result is a policy that trains in minutes, requires as few as 20 demonstrations, achieves a 97% stabilisation success rate with 40 training trajectories and 100% with 80 [1], and runs as a single matrix multiply at inference time with no GPU and no simulator.

As a first step toward extending this framework to multi-axis systems, this work considers a cartpole mounted on a linear axis driven by a stepper motor. This platform preserves

the key challenge of vision-only feedback while introducing a control output that is a discrete step command $u_k \in \mathbb{Z}$ rather than a continuous torque. Stepper-driven linear axes are the building blocks of low-cost robot arms, making the cartpole a natural stepping stone toward the broader goal of pixels-to-steps control for multi-axis systems. The present study evaluates the observer in simulation using the Gymnasium environment [2] as a testbed, with the application scenario shown in Fig. 1.

II. PROBLEM DEFINITION

A. State Variables and Measurements

Following the notation of Lee *et al.* [1], the discrete-time cartpole state at time step k is

$$\mathbf{x}_k = [x_k \quad \dot{x}_k \quad \theta_k \quad \dot{\theta}_k]^\top \in \mathbb{R}^4, \quad (1)$$

where x [m] is the signed horizontal cart displacement from the track centre, $\dot{x}$ [m/s] the cart velocity, θ [rad] the pole angle from the upright vertical (positive counter-clockwise), and $\dot{\theta}$ [rad/s] the pole angular rate.

The sole sensor measurement is a binary grayscale camera frame

$$\mathbf{y}_k^p \in \{0, 255\}^{W \times H}, \quad W = 600, H = 117 \text{ px}, \quad (2)$$

captured at 60 Hz. Frames are binarised (pole pixels white on black background) and zero-padded to a canonical 117×600 geometry. The control input $u_k \in \mathbb{R}$ is known because it is commanded by the controller. The key geometric relationship exploited by the pixel-based observer is that the pole-tip pixel column u^* satisfies, to first order,

$$u^* \approx \frac{W}{2x_{\max}} x + \frac{Wl}{2x_{\max}} \theta, \quad (3)$$

making the raw pixel vector an approximately linear function of (x, θ) —the geometric basis for the LLS approach of [1].

B. Control Objective

An LQR state-feedback gain $\mathbf{K} \in \mathbb{R}^{1 \times 4}$ is designed on nominal plant parameters. The controller is

$$u_k = -\mathbf{K} \hat{\mathbf{x}}_k, \quad \mathbf{K} = [-31.5 \quad -79.8 \quad -970.8 \quad -109.5], \quad (4)$$

where $\hat{\mathbf{x}}_k$ is the observer estimate. The observer's task is to produce estimates accurate enough for (4) to stabilise the pole.

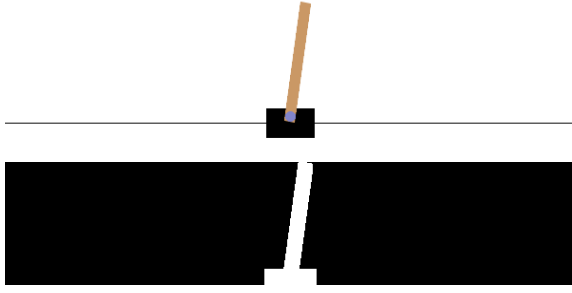


Fig. 1: *Top*: simulation frame at $\theta_0 = +8^\circ$. *Bottom*: corresponding binarised, zero-padded 117×600 px observation $\mathbf{y}^p$ supplied to the observer. The pole silhouette is the sole sensor signal; no encoder or IMU is available.

III. DATA AND DATA SOURCE

A. Simulation Environment

All data are collected in the Gymnasium CartPole-v1 environment [2]. Nominal plant parameters are $m_c = 1.0$ kg, $m_p = 0.1$ kg, $l = 0.5$ m, $g = 9.8$ m/s², $\tau = \frac{1}{60}$ s, and actuator saturation $|u| \leq 10$ N.

To assess robustness, all evaluation rollouts use a mismatched true plant: $m_p \leftarrow 1.1 m_p$ and $l \leftarrow 0.9 l$, mimicking the 5% parametric error used in [1]. The observer is unaware of the mismatch.

B. Teacher Demonstrations

A privileged LQR teacher—with access to the full true state—collects stabilising trajectories. Following the protocol of Lee *et al.* [1], each demonstration begins at rest ($x_0 = \dot{x}_0 = \dot{\theta}_0 = 0$) with a fixed initial pole angle $\theta_0 \in \{\pm 0.5^\circ, \pm 1.0^\circ, \dots, \pm 12.0^\circ\}$, giving 150 unique trajectories each running for 150 steps (2.5 s at 60 Hz).

Frames are rendered at native resolution and zero-padded (top and symmetric sides) to a canonical 117×600 px geometry. Each frame is binarised by thresholding to isolate the pole silhouette. The resulting dataset contains **35,928** (frame, state) pairs. Fig. 2 shows example binarised frames across the angle range.

C. Ridge Regression Fit Quality

A Ridge regressor ($\alpha = 0.1$) trained on the 150-demo dataset to predict pole angle from the flattened pixel vector yields $R^2 = 0.986$, RMSE = 0.115° . Only 12,952 of the 70,200 pixel coefficients are nonzero, confirming the sparse spatial support concentrated on the pole-tip region observed by Lee *et al.* [1] in their Fig. 4.

IV. METHOD OF STATE ESTIMATION

A. Pixel-Based Luenberger Observer (Lee *et al.*)

Lee *et al.* [1] parameterise the observer as

$$\hat{\mathbf{x}}_{k+1} = A_L \hat{\mathbf{x}}_k + B_L u_k + \mathbf{L} \mathbf{y}_{k+1}^p + \mathbf{d}, \quad (5)$$

where $A_L \in \mathbb{R}^{4 \times 4}$, $B_L \in \mathbb{R}^{4 \times 1}$, $\mathbf{L} \in \mathbb{R}^{4 \times WH}$, and $\mathbf{d} \in \mathbb{R}^4$ are all *learned* jointly. Defining the parameter matrix $\Theta =$

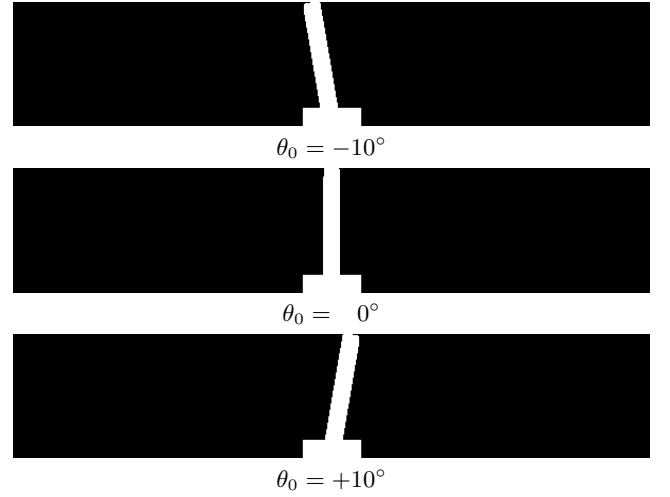


Fig. 2: Binarised 117×600 px training frames at three initial angles. The pole silhouette shifts laterally and tilts as x and θ change, providing the approximately linear pixel signal exploited by the Ridge regressor.

$[A_L, B_L, \mathbf{L}, \mathbf{d}]$, teacher demonstration data are gathered into matrices $\hat{X}_{2:N}$ (predicted next states) and Φ (current states, inputs, pixel frames, bias column), then the regularised LLS problem is solved:

$$\min_{\Theta} \|\Theta \Phi - \hat{X}_{2:N}\|_F^2 + \lambda \|\mathbf{L}\|_1, \quad (6)$$

with an optional LMI constraint enforcing closed-loop stability. Crucially, *all four rows* of $\mathbf{L}$ are non-zero after fitting, meaning the observer extracts pixel information relevant to x , $\dot{x}$, θ , and $\dot{\theta}$ simultaneously.

B. Hybrid Observer (This Work)

The implementation here fixes A_L, B_L to the nominal linearised dynamics (not learned) and sets $\mathbf{L} = \mathbf{0}$, $\mathbf{d} = \mathbf{0}$. Instead of solving (6), a Ridge regression model is trained separately to map pixels to pole angle:

$$\hat{\theta}_{k+1}^{\text{px}} = \mathbf{c}_\theta^\top \bar{\mathbf{y}}_{k+1}^p + b_\theta, \quad (7)$$

where $\bar{\mathbf{y}}^p \in [0, 1]^{WH}$ is the normalised pixel vector. The predictor step is

$$\hat{\mathbf{x}}_{k+1}^- = A_L \hat{\mathbf{x}}_k + B_L u_k, \quad (8)$$

and the angle component is corrected via a convex blend:

$$\hat{\mathbf{x}}_{k+1}[2] \leftarrow (1 - w_\theta) \hat{\mathbf{x}}_{k+1}^-[2] + w_\theta \hat{\theta}_{k+1}^{\text{px}}, \quad w_\theta = 0.7. \quad (9)$$

The remaining components ($x, \dot{x}, \dot{\theta}$) retain the model prediction. This is equivalent to (5) with

$$\mathbf{L}_{\text{eff}}[i, :] = \begin{cases} w_\theta \mathbf{c}_\theta^\top & i = 2 (\theta), \\ \mathbf{0}^\top & \text{otherwise.} \end{cases} \quad (10)$$

TABLE I: Observer Design Comparison

	Lee et al. [1]	This work
A_L, B_L	Learned (LLS)	Fixed (linearised)
$\mathbf{L}$	Learned, all rows	Effective (1 row)
$\mathbf{d}$	Learned	Zero
Training	Joint LLS	Separate Ridge
Stability constraint	LMI (optional)	N/A
States corrected	All 4	θ only

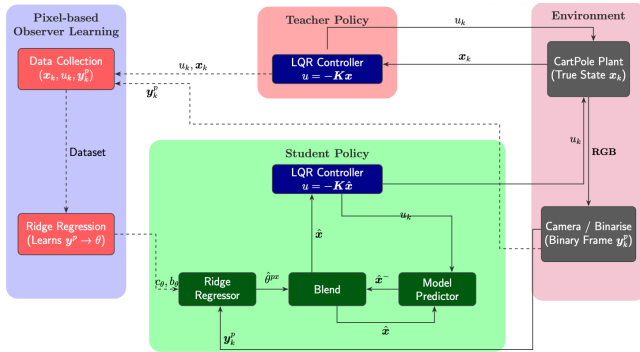


Fig. 3: System block diagram. All signals are updated at 60 Hz.

C. Comparison to Prior Work

Table I contrasts the hybrid observer against the full learned observer of Lee *et al.* [1].

Unlike a Kalman filter, neither observer computes an error covariance or updates the gain online. The blend weight w_θ plays the role of a fixed Kalman gain for the angle channel. Compared to a particle filter, no Monte Carlo sampling is required; the entire update is a single matrix-vector product (8) and one dot product (7).

D. System Block Diagram

Fig. 3 shows the full signal flow. At each 60 Hz timestep: (1) the camera image is binarised; (2) the Ridge model produces $\hat{\theta}^{\text{px}}$; (3) the model predicts $\hat{x}^-$; (4) the blend replaces the angle component; (5) the updated $\hat{x}$ drives the LQR controller.

V. EXPERIMENTS AND RESULTS

A. Evaluation Protocol

Following Lee *et al.* [1], performance is evaluated across 100 random initial conditions $\theta_0 \sim \mathcal{U}(-12^\circ, +12^\circ)$ (with $x_0 = \dot{x}_0 = \dot{\theta}_0 = 0$) per training-set size. Each rollout runs for 150 steps (2.5 s) on the mismatched plant. A trial fails early if $|x| > 2.4$ m or $|\theta| > 12^\circ$.

B. Performance Metric

The stabilisation error at the evaluation deadline is

$$e_{\text{stab}} = \|\hat{\mathbf{x}}_{150}\|_2, \quad (11)$$

the Euclidean norm of the estimated state at step 150. A trial is a *paper success* if both criteria hold at step 150:

$$|x_{150}| < 0.176 \text{ m} \quad \text{and} \quad |\theta_{150}| < 2^\circ. \quad (12)$$

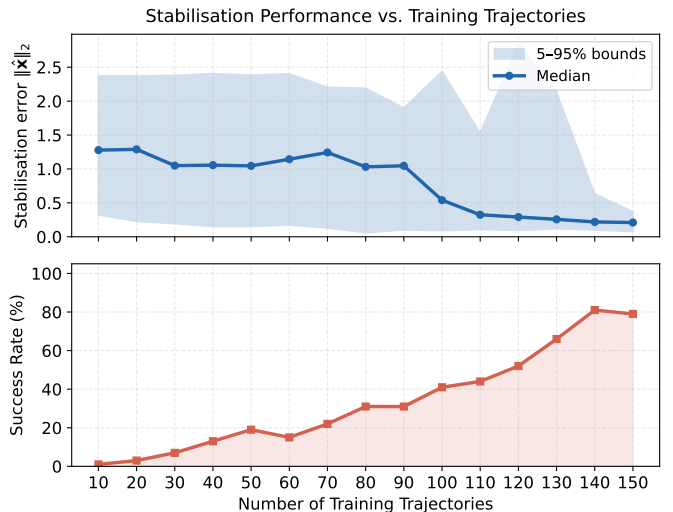


Fig. 4: Stabilisation performance vs. training-set size (100 trials per point, mismatched plant). *Top*: median e_{stab} (Eq. (11)) with 5th–95th percentile band. *Bottom*: paper-success rate (Eq. (12)). Both curves plateau near 150 demos; the main gain occurs between 100–140 demos where the tail of the error distribution collapses.

The threshold of 0.176 m is half the pole length ($0.5 \text{ m} \times 0.352$), matching the definition in [1].

C. Stabilisation Performance vs. Training Data

Fig. 4 reproduces the sweep experiment of Lee *et al.* [1] (their Fig. 3) using the hybrid observer. At 40 demos the success rate is only 13%, compared to 97% in the original paper. This gap arises directly from the structural difference in Table I: the full observer of [1] learns non-zero $\mathbf{L}$ rows for *all four states* from pixels, correcting x , $\dot{x}$, and $\dot{\theta}$ as well as θ . The hybrid observer corrects only θ , so it requires far more data to achieve stable cart regulation through the LQR feedback loop. By 140–150 demos, the angle regressor is accurate enough that the model-predicted x error remains within the controller’s tolerance.

D. Single-Rollout Time Histories

Fig. 5 corresponds to Fig. 5b of Lee *et al.* [1] for a single stabilisation rollout. The pole angle estimate (middle panel) tracks the true angle tightly, while the cart position estimate (top panel) drifts because only the open-loop model propagates x . Despite this estimation error, the applied control force (bottom panel) closely follows the teacher-anticipated command, confirming that angle correction alone is sufficient for closed-loop stabilisation within the 2.5 s window.

E. Effective Gain and Spatial Interpretation

The effective gain $\mathbf{L}_{\text{eff}}$ defined by (10) has non-zero entries only in its third row (pole angle). These coefficients, when reshaped to the frame geometry, concentrate near the pole tip—consistent with the interpretable sparse structure reported in

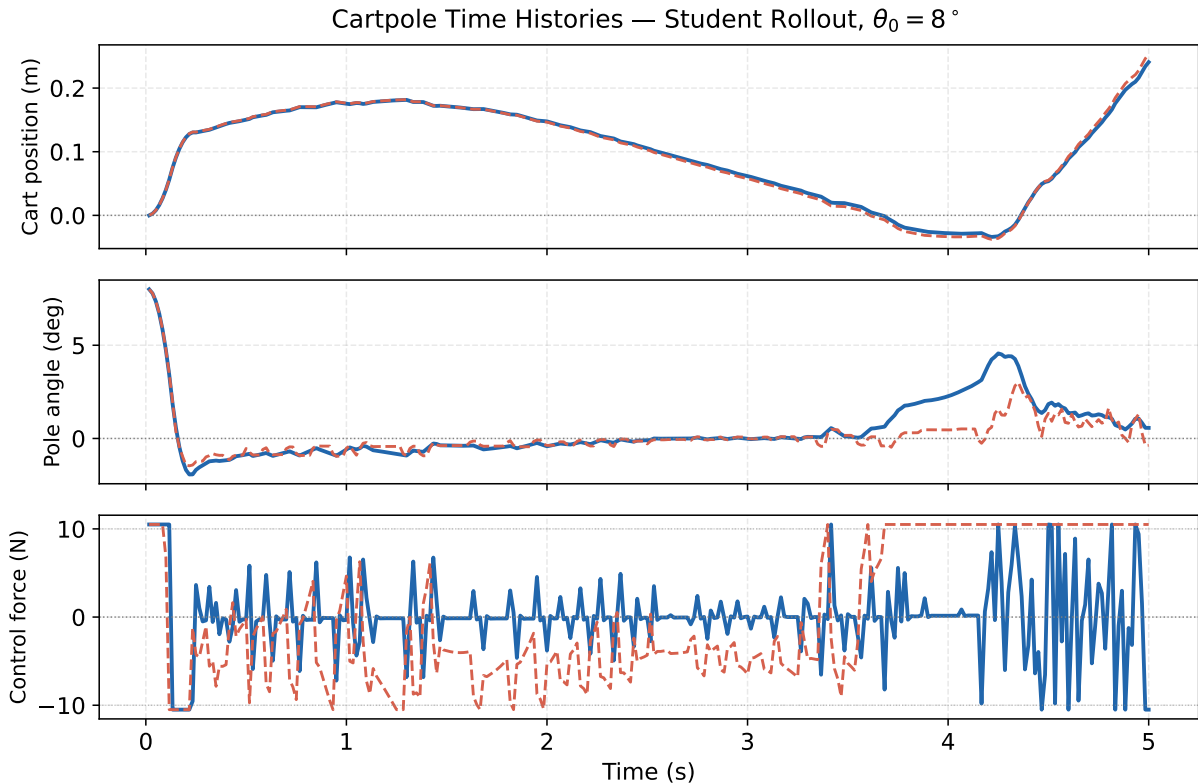


Fig. 5: Time histories for a single rollout at $\theta_0 = 8^\circ$ (150-demo observer, mismatched plant). *Top*: cart position; *Middle*: pole angle. True state (solid) vs. observer estimate (dashed). *Bottom*: applied control force vs. teacher anticipated command. The pole angle is tracked closely; cart position error accumulates but remains within the LQR basin.

TABLE II: Observer Performance at 150 Training Demos (100 Evaluation Trials, Mismatched Plant)

Metric	This work	Lee et al. [1]
Training demos for success	150	40–80
Student survival (150 demos)	100%	100%
Paper success (150 demos)	79%	$\geq 97\%$
e_{stab} median	0.210	—
e_{stab} 95th pct	0.372	—
Ridge R^2 (θ)	0.986	—
Ridge RMSE (θ)	0.115°	—

[1] (their Fig. 4, θ row). In contrast, [1] also reports non-zero cart and velocity rows (Fig. 4a, 4b, 4d), which the fixed $\mathbf{L} = \mathbf{0}$ structure of the hybrid observer cannot reproduce.

F. Quantitative Summary

VI. CONCLUSIONS

- 1) **The pixels-to-torques framework is reproducible with lightweight tools.** A Ridge regression model trained on raw binary frames replicates the angle-estimation accuracy ($R^2 > 0.98$, RMSE $< 0.12^\circ$) reported by Lee *et al.* [1] without requiring a full LLS solve or stability-constrained optimisation.

- 2) **Correcting only θ requires substantially more data.** At 40 demos, the success rate is 13% versus 97% in [1]. The full joint LLS observer learns pixel corrections for all four states simultaneously, giving it far better data efficiency because cart and velocity errors are damped directly rather than only indirectly through the LQR loop.
- 3) **Cart-position estimation is the key open problem.** With no direct pixel signal for x or the velocities, drift accumulates beyond the 2.5 s evaluation window. Adding a second Ridge regressor for x (feasible given that the pole-tip column encodes x , as in (3)) is the natural next step toward matching full-observer performance.
- 4) **The method is interpretable and connects to classical theory.** The effective gain (10) makes the hybrid observer a special case of (5), enabling the same closed-loop stability analysis used in [1]. The sparse, spatially localised pixel coefficients confirm the physical interpretation: the observer learns to look where the geometry says it should.

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